Mentor Call Preparation

Eton College iGEM 2026 • Dry Lab Meeting Agenda

Aaran Nihalani • 3 July 2026 • 21:00 BST

Agenda Item 1: Why Proteases, Why MMP14, and Why Cancer

Why activity-based detection over presence-based detection

Standard ELISA and antibody assays measure *how much* of a protein is present. For proteases, disease relevance lies in *enzymatic activity*: an MMP14 molecule that is present but inhibited by TIMPs contributes nothing to tumour invasion; an active molecule does. ELISA cannot distinguish these two states. Our translator reads activity directly, producing a signal only when the protease is catalytically competent.

Wider validation from our OpenAlex literature survey (1,293 papers, 2019–2026): the subfield is growing at 140 to 241 papers per year; cancer is the leading application area; and the fastest-growing detection methods are engineered reporters (exactly our approach), not classical immunoassays.

Why MMP14 specifically

MMP14 (MT1-MMP) is overexpressed and catalytically active in the tumour microenvironment (TME) across breast, colorectal, gastric and pancreatic cancers. Its primary functions in cancer progression are:

- Pericellular collagen IV and fibronectin degradation, enabling local invasion
- Activation of proMMP-2, amplifying the MMP cascade
- Direct shedding of CD44 and other surface receptors, promoting migration and angiogenesis

Serum MMP14 concentrations from published literature provide a clear, quantitative disease signal:

Group	**[MMP14] serum**	**Approx. pM**
Healthy controls	∼0.62 ng/mL	∼9 pM
Cancer patients (range)	2.1–4.6 ng/mL	32–70 pM
Fold change	3.4× to 7.4×	

Sources: Laudanski 2010 (breast cancer, PMID 20529824); Peltonen et al. 2018 (gastric cancer, PMID 30532247).

Why MMP14 and not other disease states

The concern is legitimate: elevated MMP14 is not exclusive to cancer. Key points:

- MMP14 is also elevated in rheumatoid arthritis, fibrosis, and atherosclerosis; MMP14 alone does not constitute a cancer-specific diagnosis
- The translator is a **modular activity-measurement device**, not a standalone diagnostic; clinical interpretation combined with other biomarkers or imaging makes the diagnosis
- Our choice of SGRSENIRTA adds a kinetic selectivity layer: MMP14 cleaves it at 79% relative efficiency vs. 4% for MMP2 and 9% for MMP9, giving at least an 8-fold MMP14 preference within the MMP family
- If stronger disease specificity is required in v2, the modular architecture supports a dual-protease panel: replacing the cleavage motif with a FAP-selective peptide, or combining MMP14 and FAP signals (both TME-enriched), would sharpen the cancer signature considerably

The answer for the mentor: MMP14 activity is a high-confidence TME marker; the translator measures it quantitatively; clinical deployment would use it as part of a panel, not in isolation.

Agenda Item 2: The Output Readout

Current leading mechanism (v4 design)

The v4 document specifies an electrochemical readout using methylene blue (MB) as the redox reporter. Mechanism:

1. The intact translator is tethered to a gold electrode via the His6 terminus
2. MB is conjugated to the S6 socket via an MB-CoA conjugate, delivered by Sfp synthase
3. In the intact (uncleaved) state, the protein holds MB within electron-transfer distance of the electrode; square-wave voltammetry (SWV) produces a characteristic faradaic peak
4. MMP14 cleaves SGRSENIRTA; the MB-bearing fragment diffuses into bulk solution, severing electron transfer; the SWV peak falls in proportion to protease activity
5. Readout: quantitative, signal-off, requiring a gold electrode and a handheld potentiostat

Output method comparison

Method	Pros	Cons	iGEM demo-ability
Electrochemical (MB)	Quantitative, PoC-compatible, no optics needed, handheld potentiostat available	Requires MB-CoA synthesis; electrode monolayer stability is a risk	High
Glucometer	Extremely cheap, globally available reader	Glucose oxidase-CoA conjugate non-trivial; narrow dynamic range	Medium
Lateral flow	Visible, no instrument required	Binary output, not quantitative; requires colloidal gold conjugation	Medium
CRISPR/Cas12a	Amplified signal, fluorescent readout	Two-step workflow; more failure points	Low
FRET (fluorescent)	Lab-proven, easy to characterise	Not PoC; requires plate reader	Low

Recommendation

Commit to electrochemical (MB / gold electrode / SWV) as the primary readout for iGEM 2026. Reasons:

- Directly supported by v4 design rationale and the Dr Devoy collaboration framework
- Quantitative and PoC-compatible, strengthening the project's real-world impact argument for judges
- Dry-lab structural modelling of linker length and electron-transfer distance has a direct, testable interface with wet-lab SWV measurements
- The second socket (ybbR) is available for a ratiometric internal control, mitigating monolayer drift risk

Retain glucometer as a backup output for a v2 demonstration if MB-CoA synthesis proves intractable before the iGEM deadline.

Agenda Item 3: Blood Volume Back-Calculation

Model and parameters

Under pseudo-first-order conditions ($[S]_0 \ll K_M$, where $K_M \sim 10$–100 μM for MMPs and $[S]_0 = 100$ nM):

$$[\text{Cleaved}] \approx \frac{k_{\text{cat}}}{K_M} \cdot [E] \cdot [S]_0 \cdot t$$

Parameter	Value	Source
k_{cat}/K_M (SGRSENIRTA + MMP14)	5.5×10^4 $\text{M}^{-1}\text{s}^{-1}$	Literature-derived (conservative)
Sensor concentration $[S]_0$	100 nM	Design parameter
Incubation time t	30 min	Design parameter
Detection threshold $[\text{Cleaved}]_{\text{min}}$	0.1 nM	Conservative (electrochemical)
Serum yield from whole blood	50%	Standard clinical assumption

From these parameters: $[\text{Cleaved}] = 9.9 \times [\text{MMP14}]_{\text{reaction}}$, giving a minimum required MMP14 concentration in the reaction of **10.1 pM**.

Predicted signal at clinical MMP14 concentrations

Group	[MMP14] serum	Cleaved after 30 min	Above threshold?
Healthy	9 pM	0.089 nM	No
Cancer (low)	32 pM	0.317 nM	Yes (3.2× threshold)
Cancer (high)	70 pM	0.693 nM	Yes (6.9× threshold)

Required blood volume by reaction size

Minimum serum fraction: $f = 10.1$ pM / $[\text{MMP14}]_{\text{serum}}$. Blood volume $\approx 2\times$ serum volume.

[MMP14] serum	Rxn vol	Serum needed	Blood needed	Collection
32 pM (cancer low)	20 μL	6.3 μL	**13 μL**	Finger-prick
32 pM	50 μL	15.8 μL	**32 μL**	Finger-prick
70 pM (cancer high)	20 μL	2.9 μL	**6 μL**	Finger-prick
70 pM	50 μL	7.2 μL	**14 μL**	Finger-prick

A standard finger-prick (20–50 μL blood) is sufficient at cancer-range MMP14 concentrations. The healthy range falls below threshold under these parameters.

Key caveats for the call:

- $k_{\text{cat}}/K_M = 5.5 \times 10^4$ $\text{M}^{-1}\text{s}^{-1}$ is a literature-derived estimate pending wet-lab measurement with our expressed construct
- Healthy-range MMP14 (9 pM) produces 0.089 nM cleaved, close to the 0.1 nM threshold; a safety margin and statistical noise analysis is a near-term dry-lab task
- Detection threshold assumes electrochemical readout; a different output modality shifts all numbers

Agenda Item 4: Model-to-Wetlab Integration Plan

Reference: Heidelberg 2025

The Heidelberg 2025 model is the stated benchmark. Their key feature: 18 construct designs proposed and ranked by model predictions, with wet-lab selection determined computationally. The judges rewarded this because the model was not decorative; it decided which constructs were built.

The question for us: which specific model outputs drive which wet-lab decisions?

Integration architecture

Stage	Dry-lab output	Wet-lab decision driven	Owner
1	Structural scorer: SASA at cleavage site, anchor geometry, k_{open}	Ranked list of linker variants to express	Aaran
2	Kinetics ODE: [Cleaved](t) vs [MMP14] curves	Incubation time and sensor concentration for assay	Archer, Aniu
3	Electrochemical geometry: MB-to-electrode distance vs linker length	Linker length selection; expected SWV peak current	Aaran
4	Sobol sensitivity: which parameter dominates the signal?	Priority order for wet-lab optimisation	Archer, Aniu
5	Wet-lab k_{cat}/K_M feeds back into Stage 2	Refines kinetics model; closes the loop	Wet lab → Dry lab

Wiki model page deliverables

Concrete components for the iGEM wiki model page:

- **Structural model:** AlphaFold3 structure of the full translator construct annotated with domain boundaries, cleavage site SASA, and ybbR/S6 anchor geometry. Outputs k_{open} as input to kinetics model.
- **Kinetics ODE model:** 4-stage cascade (cleavage, conformational release, signal transduction, readout). RK45 integration. Produces [Cleaved](t) and endpoint signal as a function of [MMP14]. Validated against wet-lab cleavage assay data.
- **Sensitivity analysis:** Sobol indices on kinetics model parameters. Identifies the rate-limiting step and the variable wet lab should prioritise.
- **Blood volume calculator:** The back-calculation from Section 3, packaged as a web tool (software prize component).
- **Electrochemical geometry model:** Electron-transfer distance as a function of linker length via Marcus theory ($k_{ET} \propto e^{-\beta d}$). Identifies optimal GGGGS repeat count placing MB within 1–3 nm of the electrode surface.

Timeline and the key interface

The dry-to-wet interface is the k_{cat}/K_M measurement. Dry lab provides wet lab with a ranked construct list from the structural scorer. Wet lab returns measured k_{cat}/K_M values. Those values update the kinetics model. The updated model re-ranks remaining constructs. This closes the loop.

The structural scorer's first output (ranked linker variants) is due end of July. Wet-lab cleavage assays begin in August. This is achievable within the iGEM timeline and maps directly onto the Heidelberg integration model.